Complete Git & GitHub Notes

1. What is Git?

Git is a version control system that helps track changes, undo mistakes, and collaborate.

2. What is GitHub?

GitHub is a cloud platform for hosting Git repositories.

3. Install Git

Download from https://git-scm.com and verify using: git --version

4. Git Basic Configuration

git config --global user.name "Your Name"

git config --global user.email "yourmail@example.com"

5. Initialize a Repository

git init

6. Basic Git Commands

git status

git add .

git commit -m "message"

7. GitHub Setup

git remote add origin

git push -u origin main

8. Cloning a Repository

git clone

9. Pulling Changes

git pull

10. Branching

git branch feature

git checkout feature

git checkout -b feature

git merge feature

11. Working With Remotes

git remote -v

git remote set-url origin

git remote remove origin

12. Authentication

Use Personal Access Token (PAT) instead of password.

13. Common Git Errors

fatal: not a git repository → run inside correct folder.

repository not found → wrong URL/login.

merge conflict → fix manually then commit.

14. Git Workflow

git pull

git add .

git commit -m "message"

git push

15. .gitignore

Use to ignore node_modules, .env, build, etc.

16. GitHub Best Practices

Commit often, meaningful messages, use branches, pull before push.

17. GitHub Desktop

Optional GUI for beginners.

18. Switching GitHub Accounts

Remove old credentials → push → login with new PAT.